

LogiFlow — Week 2 Progress Report

Candidate: Aryan Ajmera

Reference Number: NFGT/HR/INT/2026/160

Week: 2 of 8 (Backend Engineering & REST APIs — Weeks 1–4)

Date: July 31, 2026

1. Tasks Completed

Task W2.1 — Warehouse & Product Management APIs

- Implemented full CRUD for Warehouse and Product: create, list (filterable, paginated), get by id, update, and soft delete
- Restricted create/update/delete to Administrator and Warehouse Manager; all roles retain GET access
- Tested all 10 endpoints in Postman with valid, invalid, and unauthorized scenarios

Task W2.2 — Inventory Management APIs

- Implemented inventory create/list/get, plus restock and deduct endpoints, each logging an ActivityLog entry
- Implemented a business rule blocking deduction below zero stock, returning a 400 on over-deduction
- Implemented auto-flagging of lowStockAlert on deduction below reorderThreshold, and auto-clearing it on restock
- Implemented GET /api/inventory/low-stock and a triggerLowStockNotification() utility that creates a Notification for the warehouse manager when stock drops below threshold
- Tested edge cases: zero/negative deduction, over-deduction, negative restock, and the full deduct → alert → notification chain

Task W2.3 — Week 2 Integration & Review

- Verified the end-to-end flow: Warehouse → Product → Inventory → Deduct → low-stock flag → Notification
- Confirmed role restrictions hold across Warehouse, Product, and Inventory endpoints
- Committed all work to the develop branch

2. Blockers Faced & Resolutions

- **Missing _id in create responses:** POST /api/warehouses, /api/products, and /api/inventory manually built their JSON responses and each omitted the document's _id, blocking downstream tests that needed those ids. Resolved by adding _id explicitly to all three response objects (createWarehouse also referenced a nonexistent warehouseId field, which was removed).
- **Low-stock notification not generating:** Deducting stock correctly flipped lowStockAlert to true, but no Notification document appeared. Traced to triggerLowStockNotification() only firing when the linked Warehouse has a managerId set — the test warehouse had none, so the step was silently skipped. Resolved by setting managerId via PUT /api/warehouses/:id.

- **PUT update appearing not to persist:** An initial managerId update did not reflect on a follow-up GET. Re-sent as a clean raw/JSON body and it persisted correctly — likely a malformed request body rather than a controller issue.

3. What Was Learned

- Why _id must be explicitly included in manually constructed response objects
- Designing safe stock-mutation logic and deriving boolean alerts from numeric thresholds
- How cross-document side effects depend on referenced fields being populated, and the risk of failing silently rather than loudly
- Debugging directly against MongoDB Atlas/Compass rather than trusting the API response alone

4. Plan for Next Week (Week 3)

- Task W3.1: Customer & Order Management APIs, including stock-validated order creation and the order status machine
- Task W3.2: Shipment Lifecycle Management APIs, with tracking events and order auto-sync
- Task W3.3: Week 3 integration testing and review